

PHY1004W: *Matter and Interactions* Course Information 2022

First semester edition (will be updated for the second semester)

PHY1004W (*Matter and Interactions*) is the calculus-based first-year course for science students who intend to proceed to higher physics courses. It is also suitable for mathematicians, astronomers, chemists, computer scientists and geologists. Together with MAM1000W it is a prerequisite for the further physics course PHY2004W, and then PHY3004W which leads to the physics major. Students will normally be expected to have passed NSC Physical Science with at least 60% and Mathematics with at least 70%. A full first-year Mathematics course such as MAM1000W (or equivalent) must have been passed or be taken concurrently. Applied Maths I is also strongly recommended.

Due to the ongoing COVID-19 pandemic, this course will be delivered partially online and partially face-to-face. The main structure of the “lectures” will be delivered through Vula lessons. However all students should be prepared to come to campus every day, although the class will be split in half, with half the class attending a live session on any on day in RW James LT4A at 10:00. The laboratory component will be delivered both in PHYLAB1 and in a do-at-home format. Tests and examinations will be administered on campus.

All students in this course are expected to have a laptop and a stable internet connection for all aspects of the course.

The course consists of approximately 120 “lectures”, and 24 tutorial or laboratory sessions. The PHY1004W course convenor is: **Prof Andy Buffler**, (andy.buffler@uct.ac.za) in the first semester.

Course outline

The course will cover ...

- Mechanics: momentum principle, conservation of energy, energy quantization, angular momentum
- Matter: atomic nature of matter, waves and particles
- Thermodynamics: statistical physics, entropy, probability theory
- Electricity: Fields and charges, potential, circuits, Gauss' Law
- Magnetism: Fields and currents, magnetic forces, Ampere's Law, Faraday's Law
- Electromagnetic radiation, waves, physical optics

... with an emphasis on ... analyses of systems based on a small number of fundamental principles; the atomic nature of matter, and the link between microscopic and macroscopic phenomena; a unified treatment of topics; and modelling of complex physical systems, including the use of numerical (computer) treatments.

Lecturers

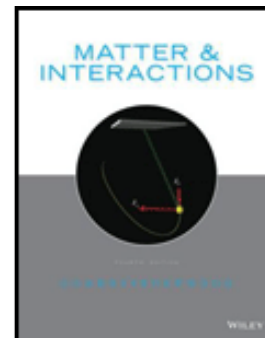
- Mechanics: Andy Buffler, Trisha Salagaram, Spencer Wheaton
- Electromagnetism: (TBA), Steve Peterson, Andre Peshier

Textbook

The prescribed textbook is *Matter and Interactions 4th Edition*, Ruth Chabay and Bruce Sherwood, John Wiley and Sons, 2015.

Lectures

All “lectures” will be delivered on the Lessons platform on Vula. Students will be expected to attend face-to-face discussions twice a week in RW James LT4A at 10:00. The class will be split into two groups.



Computing

The numerical modelling of physical systems using a computer is a central part of the course. Instruction will be provided using the computer language Python (version 3.8.1) together with the visual module. This [VPython](#) package is used in the textbook. All students are required to have their own laptop with Python and VPython installed. There is a separate document dealing with installing Python and VPython.

Weekly problem sets

Each Friday morning a new weekly problem set will appear on Vula. Students are to work through all the problems (and are strongly encouraged to attempt the extra, textbook problems listed at the bottom of the sheet as well) by the end of the next week. (Students may consult with each other and approach the course tutor for help if necessary.) Full solutions to these problems must be handed in by each student on the Vula assignments tab set up for WPSs. Your WPS will be graded between 0 and 5 although your solutions will not be thoroughly checked. Worked solutions to the questions will be published on Vula for you to check your own work. Marks obtained for these weekly problem sets will contribute 5% towards the final course mark. *Note: These weekly problem sets and the tutorials are a good indicator of the type and standard of questions which can be expected in tests and exams.*

Course tutors

The course tutors for the course are **Erin Jarvie** and **Rachel Catzel** who will deal with the weekly problem sets and be available at certain times for consultation online.

Communication

Notices, schedule changes, and other course material will be posted on the PHY1004W Vula site. You will also receive email correspondence from time to time, so check your UCT email regularly. The Vula chat room may be useful.

Laboratory

Dr Trisha Salagaram (trisha.salagaram@uct.ac.za) is the coordinator of the PHY1004W laboratory in the first semester. Laboratory sessions for PHY1004W students take place on Tuesday afternoons from 14h00 either in PHYLAB1 or at home (a schedule will be published). Certain laboratory reports will be designated as writing exercises, and particular emphasis will be placed in these on writing skills. There is a separate handout regarding laboratories.

Tutorials

Approximately once every four weeks, students will partake in a tutorial session instead of a practical. A selection of about four problems will be assigned. More details will follow later.

Assessment

PHY1004W is assessed as shown below:

6	Class (theory) tests	20%
2	Laboratory tests	10%
24	Weekly problem sets	5%
many	Laboratory reports	15%
1	June Examination (2 hours)	25%
1	November Examination (2 hours)	25%

You can check your up-to-date grades for the course on Vula gradebook.

Duly performed (DP) requirements

A student will be regarded as having “duly performed” the work of the course, and thus qualify to write the final examination, if he/she has met the DP requirements for this course. DP certificates may be withheld from students who fail to meet these minimum requirements. Students who are not awarded DP certificates will not be permitted to write the final examination in November. The DP list will be published no later than one calendar week before the last teaching day of the course, and all grades recorded on that day will be used to consider the DP status of each student. Grades recorded after this date will be used in cases of appeal when a DP is not awarded. An appeal against a DP not being awarded is first made to the course convener, and thereafter potentially to the Head of Department (by email). The following are the **DP requirements for this course**:

1. A minimum of 35% overall for the coursework component of the course (as determined one week before the last teaching day of the course).
2. Participation in all class tests. Students missing a test for medical reasons will be required to write a make-up within three days of returning from illness, in consultation with the course convener.
3. A minimum of 50% for the laboratory component of the course.

Attendance and exemptions

Participation in all practicals, tutorials, tests and examinations is compulsory. All students are expected to attempt all practical and computational laboratories for the course, and complete all laboratory reports and laboratory tests, and attend all tutorials. If you are ill and miss any grade-carrying activity, then a medical certificate from a registered medical practitioner needs to be emailed to Jill Patel (jill.patel@uct.ac.za) within 2 days of returning from illness, and a short form will need to be completed (available on the PHY1004W Vula site). You are also required to email the course convener indicating the activities you have missed. Students missing a test due to illness will be asked by the course convener to write a make-up test within a few days. An application for exemption from laboratory activities for students who are repeating the course must be made using the special form for this purpose (available on the PHY1004W Vula site) and emailed to Mr. Mark Christians in the first-year laboratory within the first two weeks of term (mark.christians@uct.ac.za).

Short leave from the course

If a student wishes to be granted an exemption or extension for a course requirement associated with a planned (future) short absence from the course, then there is a form to complete (available on the course Vula site). This form needs to be emailed to Jill Patel (jill.patel@uct.ac.za) at least 3 working days prior to the period in question. Irreversible plans (such as flight bookings) must not be made before approval of leave is granted. Completion of the form is not required for medical certificates obtained on the day of unplanned illness.

Faculty of Science Code of Honour

We will be asking all of you to agree to this Code of Honour for the course. We will assume that everything you do for formal assessment will be your own work and that you did not solicit help from others.

The University of Cape Town's official statement of general rules and policies, including a statement of values and expected student and staff conduct, can be found here:

http://www.students.uct.ac.za/sites/default/files/image_tool/images/434/study/handbooks/2020/UCT_Handbook_3_2020.pdf

All staff and students in the Faculty of Science are required to familiarise themselves with UCT's rules and policies.

The Faculty of Science takes UCT's statement of values seriously. Among other things, every member of the Faculty of Science – student and staff – is expected to strive for personal and academic integrity; to be respectful and tolerant towards others; to honour the rights, personal choices and property of others; to take responsibility for their behaviour individually and in groups; and generally to act as a responsible citizen in our academic community and beyond. As stated in the above-mentioned document: "Truth, fairness, consistency and integrity in both academic and other work, and in all personal and institutional relationships" are key elements of UCT's academic values.

We are currently in a global crisis and recognise that distance learning will be a challenge for many students. Nonetheless, we must strive to uphold these values. We ask students to commit to this Code of Honour (i.e. doing what is morally right) when completing tasks remotely: all assignments and tests must be a student's own work, with no input from others (unless otherwise specified), and no work can be plagiarised.

Students may report to their course convener or Science Faculty if they believe that other UCT students have breached this code. The Faculty of Science commits to supporting our students in this difficult time and requests their full commitment to the integrity of the academic project.

Please commit to this by completing the "Code of Honour" Quiz on the PHY1004W Vula site.

UCT Physics Department

PHY1004W “Lecture” schedule 2022

Week	Date	Mon	Tue	Wed	Thu	Fri
7	Feb 14-18	Mech1	Mech1	Mech1	Mech1	Mech1
8	Feb 21-25	Mech1	Mech1	Mech1	Mech1	Mech1
9	Feb 28- Mar 4	Mech1	Mech1	Mech1	Mech1	Mech1
10	Mar 7-11	Mech1	Mech1	Mech1	Mech1	Mech1
11	Mar 14-18	[test day]	Test 1	Mech2	Mech2	Mech2
12	Mar 21-25	[Hum Rights]	Mech2	Mech2	Mech2	Mech2
13	Mar 28 – Apr 1	vac	vac	vac	vac	vac
14	Apr 4-8	VPython	VPython	VPython	Mech2	Mech2
15	Apr 11-15	Mech2	Mech2	Mech2	Mech2	[Easter]
16	Apr 18-22	[Easter]	Mech2	Mech2	Mech2	Mech2
17	Apr 25-29	Mech2	Mech2	[Freedom]	Mech3	Test 2
18	May 2-6	[Workers]	Mech3	Mech3	Mech3	Mech3
19	May 9-13	Mech3	Mech3	Mech3	Mech3	Mech3
20	May 16-20	Mech3	Mech3	Mech3	Mech3	Test 3
21	May 23-27	swot	swot	exams	exams	exams
22	May 30 ...	More exams				
			... then ...	... vacation		
30	Jul 25-29	EM1	EM1	EM1	EM1	EM1
31	Aug 1-5	EM1	EM1	EM1	EM1	EM1
32	Aug 8-12	EM1	[Women]	EM1	EM1	EM1
33	Aug 15-19	EM1	EM1	EM1	EM1	EM1
34	Aug 22-26	EM2	EM2	EM2	EM2	Test 4
35	Aug 29 – Sep 2	EM2	EM2	EM2	EM2	EM2
36	Sep 5-9	vac	vac	vac	vac	vac
37	Sep 12-16	EM2	EM2	EM2	EM2	EM2
38	Sep 19-23	EM2	EM2	EM2	EM2	EM2
39	Sep 26-30	EM2	EM2	EM3	EM3	Test 5
40	Oct 3-7	EM3	EM3	EM3	EM3	EM3
41	Oct 10-14	EM3	EM3	EM3	EM3	EM3
42	Oct 24-28	Test 6	swot	swot	exams	exams
43	Oct 31 ...	More exams				

AB	TS	SMW	(TBA)	SP	AP
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